

Hyunjin Choi (최현진)

I508C Sangmyung University, 31 Sangmyungdae-gil, Cheonan, Korea

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EDUCATION

Sogang University Ph.D. Department of Mechanical Engineering Dissertation Title: Assistance and Rehabilitation of People with Paretic Gait Using a Torque-controlled Wearable Robot Advisor: Kyoungchul Kong	2019
Sogang University B.S. Department of Physics (Magna Cum Laude) B.E. Department of Economics	2012

PROFESSIONAL EXPERIENCE

Sangmyung University Assistant Professor Department of Human Intelligence Robot Engineering	2020 - present
Angel Robotics Senior and Junior Researcher Robot ICT R&D Team Leader (Robot SW, UX/UI Design, Circuit Design)	2017 - 2020
University Paris-Est Créteil (UPEC) Visiting Researcher Laboratoire Images, Signaux et Systèmes Intelligents (LiSSi)	Summer 2013, 2014, 2015
Sogang University (Industry-University Cooperation Foundation) Student Researcher EXO Lab (formerly RSC Lab)	2012 - 2016

FELLOWSHIPS AND AWARDS

Gold and Bronze Medal in Powered Exoskeleton Race Cybathlon Team Angel Robotics	2020
Young Researcher Award (우수신진연구자상) Institute of Control, Robotics and Systems (제어로봇시스템학회)	2019
Top 10 Student Project Worldwide National Instruments	2017
Top 10 Finalist UAE AI & Robotics for Good Project: Angelegs	2016
Global Ph.D. Fellowship National Research Foundation (NRF), Korea Project: Study on a mechatronic rehabilitation system incorporating functional electrical stimulation and a robotic exoskeleton	2012 - 2016

PUBLICATIONS

Journal Papers

† indicates corresponding author(s)

- [1] S. An, J. Feng, E. Song, K. Kong, J. Kim, and **H. Choi**[†]. “High-Accuracy Hand Gesture Recognition on the Wrist Tendon Group Using Pneumatic Mechanomyography (pMMG)”. In: *IEEE Transactions on Industrial Informatics* 20.2 (2023), pp. 1550–1561.
- [2] H. Kim and **H. Choi**[†]. “Heel Trajectory Analysis Method of Walking using a Wearable Sensor”. In: *The Journal of the Korea Institute of Electronic Communication Sciences* 18.4 (2023), pp. 731–736.
- [3] J. Seong and **H. Choi**[†]. “A Wearable Glove System for Rehabilitation of Finger Injured Patients”. In: *KROS* 18.2 (2023), pp. 379–386.

- [4] H. Choi, J. Yeo, J. Seong, and **H. Choi**[†]. “Development of Low-cost 3D Printing Bi-axial Pressure Sensor”. In: *KROS* 17.2 (2022), pp. 152–158.
- [5] **H. Choi**[†]. “Assistance of a person with muscular weakness using a joint-torque-assisting exoskeletal robot”. In: *Applied Sciences* 11.7 (2021), p. 3114.
- [6] K. Kong[†], J. Choi, K.-W. Park, J. Park, D.-H. Lee, E. Song, B. Na, S. Jeon, T. Kim, **H. Choi**, H. Woo, J.-H. Lee, B. Kim, and D.-w. Rha. “The History and Future of the WalkON Suit: A Powered Exoskeleton for People With Disabilities”. In: *IEEE Industrial Electronics Magazine* (2021), pp. 2–15. DOI: 10.1109/MIE.2021.3113267.
- [7] **H. Choi** and K. Kong[†]. “A soft three-axis force sensor based on radially symmetric pneumatic chambers”. In: *IEEE Sensors Journal* 19.13 (2019), pp. 5229–5238.
- [8] **H. Choi**[†], B. Na, J. Lee, and K. Kong. “A user interface system with see-through display for WalkON suit: a powered exoskeleton for complete paraplegics”. In: *Applied Sciences* 8.11 (2018), p. 2287.
- [9] **H. Choi**, S. Oh, and K. Kong[†]. “Control of a robotic manipulator in the polar coordinate system using a biarticular actuation mechanism”. In: *International Journal of Control, Automation and Systems* 14.4 (2016), pp. 1095–1105.
- [10] B. Na, **H. Choi**, and K. Kong[†]. “Design of a direct-driven linear actuator for a high-speed quadruped robot, cheetaroid-I”. In: *IEEE/ASME Transactions on Mechatronics* 20.2 (2014), pp. 924–933.

Refereed Conference Paper

- [11] T. Kim, E. Song, S. An, **H. Choi**, and K. Kong[†]. “Leaf-Inspired FSR Array and Insole-Type Sensor Module for Mobile Three-Dimensional Ground Reaction Force Estimation”. In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2024.
- [12] S. An, E. Song, **H. Choi**[†], and K. Kong[†]. “Measurement of Force Myography in Level Gait and Stair Gait”. In: *IFAC-PapersOnLine* 55.27 (2022). th IFAC Symposium on Mechatronic Systems, pp. 484–489.
- [13] E. Song, S. An, **H. Choi**[†], and K. Kong. “Soft Sensor Unit for Measurement of Large Scale Multi-axis Force”. In: *2022 13th Asian Control Conference (ASCC)*. IEEE. 2022, pp. 1290–1295.
- [14] E. Song, J. Seong, and **H. Choi**[†]. “Design of Multi-channel Load Cell Amplifier for 3-axis Ground Reaction Force Measurement of Exoskeletons”. In: *The 22nd International Conference on Control, Automation, and Systems (ICCAS)*. ICROS. 2022.
- [15] S. An, **H. Choi**, and K. Kong[†]. “Development of Wireless Pneumatic Myography Sensor for Real-time Muscle Contraction Measurement”. In: *2021 21st International Conference on Control, Automation and Systems (ICCAS)*. ICROS. 2021, pp. 1895–1900.
- [16] **H. Choi**, B. Na, D. Park, B. Yoo, S. Kim, D. Kim, K. Kong, Y. Chang, and D. Pha. “Effect of a Torque-controlled Wearable Robotics Device in a Child with Incomplete Paraplegia due to Spina Bifida”. In: *13th International Society of Physical and Rehabilitation Medicine (ISPRM) World Congress*. 2019.
- [17] **H. Choi**[†], B. Na, S. Kim, J. Lee, H. Kim, D. Kim, D. Cho, J. Kim, S. Shin, D.-W. Rha, et al. “Angel-suit: A modularized lower-limb wearable robot for assistance of people with partially impaired walking ability”. In: *2019 Wearable Robotics Association Conference (WearRAcon)*. IEEE. 2019, pp. 51–56.
- [18] **H. Choi**, J. Lee, and K. Kong[†]. “A human-robot interface system for walkon suit: A powered exoskeleton for complete paraplegics”. In: *IECON 2018-44th Annual Conference of the IEEE Industrial Electronics Society*. IEEE. 2018, pp. 5057–5061.
- [19] **H. Choi**, P.-G. Jung, K. Jung, and K. Kong[†]. “Design and fabrication of a soft three-axis force sensor based on radially symmetric pneumatic chambers”. In: *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. 2017, pp. 5519–5524.
- [20] **H. Choi**, K. Kim, P.-G. Jung, B. Na, D.-w. Rha, K. Jung, and K. Kong[†]. “Ankle joint moment estimation based on smart shoes”. In: *20th IFAC World Congress*. Vol. 50. 1. Elsevier, 2017, pp. 1366–1371.
- [21] J. Choi, **H. Choi**, K. Kong[†], and D. J. Hyun. “An Adaptive Disturbance Observer for Precision Control of Time-Varying Systems”. In: *IFAC-PapersOnLine* 49.21 (2016). 7th IFAC Symposium on Mechatronic Systems, pp. 240–245. ISSN: 2405-8963.
- [22] **H. Choi**, S. Oh, and K. Kong[†]. “State estimation and position control of a robotic manipulator with a biarticular actuation mechanism”. In: *IECON 2015 - 41st Annual Conference of the IEEE Industrial Electronics Society*. 2015, pp. 3503–3507.
- [23] **H. Choi**, S. Oh, and K. Kong[†]. “Design of a biarticular robotic manipulator and its control in the rotating coordinate system”. In: *2014 IEEE/ASME International Conference on Advanced Intelligent Mechatronics*. 2014, pp. 888–891.

- [24] B. Na, **H. Choi**, and K. Kong[†]. “Design of a direct-driven linear actuator for development of a cheetaroid robot”. In: *2013 IEEE International Conference on Robotics and Automation*. 2013, pp. 4023–4028.
- [25] B. Na, **H. Choi**, and K. Kong[†]. “Design of an Actuation System for a High-Speed Quadruped Robot, Cheetaroid-I”. In: *IFAC-PapersOnLine* 46.5 (2013). 6th IFAC Symposium on Mechatronic Systems, pp. 165–169. ISSN: 1474-6670.

COURSES TAUGHT AT SANGMYUNG UNIVERSITY

Signals and Systems	Fall 2020, 2021, 2022, 2023
Robot Operating System (ROS)	Fall 2020, 2021, 2022, 2023, 2024
Human-Robot Interface Engineering	Fall 2020, 2021, 2022, 2023
Control Engineering and Design	Spring 2021, 2022, 2023
Capstone Design I	Spring 2021, 2022, 2023, 2024
Probability and Statistics	Spring 2021, 2022
Healthcare and Rehabilitation Robotics	Spring 2023, 2024

PATENTS

Overshoes type device for measuring ground reaction force	Korean Patent No.10-2219037 (2021.02.17)
Driving unit of multi-joint robot	Korean Patent No.10-2244673 (2021.04.20)
Controlling method of lower body assisting robot	Korean Patent No.10-2162485 (2020.09.25)
Ground reaction force sensor module	Korean Patent No.10-2095439 (2020.03.25)
Multi-axis force sensor and multi-axis force sensing system	Korean Patent No.102041710 (2019.10.31)
Crutch for wearable robot system	Korean Patent Pending No.10-2022-0064912 (2022)
Method of using wearable assisting device	Korean Patent Pending No.10-2022-0014421 (2022)